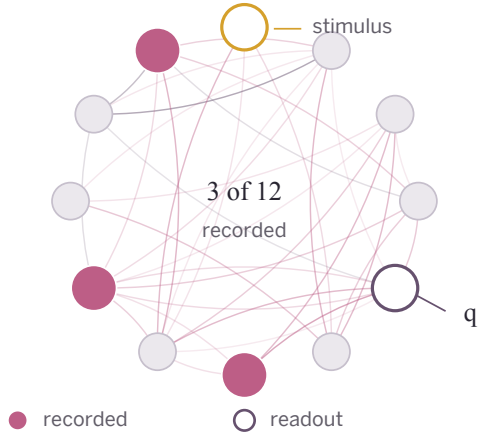
Predictive Risk and Recording Decisions in Partially Observed Neural Networks

SIMULATION STUDY

Limited recordings

Known connectivity; uncertain gains and biases



J5000 example; rose + / plum connections

Stimulus-response target



$$q = \int [r_{\text{stim}}(t) - r_{\text{control}}(t)]\, dt$$

100-unit rank-two extension

8 independent networks; exploratory

4/8 settings favour prior averaging;

one point estimate reverses sign.

Recording design

40 independent networks



Identical-set illustration

Identical recording sets

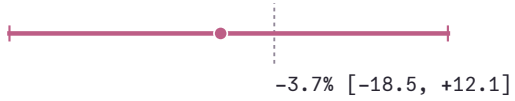
82/120 single-response designs

26/40 shared designs

Response error after MAP fitting

Same recording budget, pool and estimator

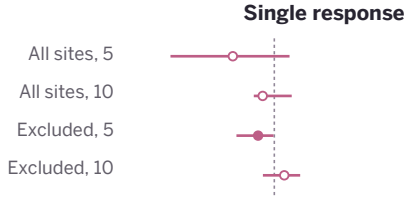
Single response



Shared three-response



97.5% network-cluster CIs; zero shown dashed



Prior-averaged minus nominal; negative favours prior averaging

Risk forecasts

40 new independent networks

Fixed recordings, data

and point prediction



MSE-forecast score

$$\log v + e^2/v$$

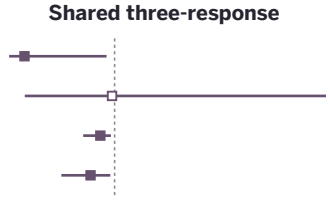
v: forecast MSE; e: response error



-0.46 [-1.16, +0.02]

95% network-cluster CI

Mean influenced by one network



Mean-MSE change (%) · unadjusted 95% CIs

Jiayi Du & Shuaizhi Cheng · Equal contribution